

Unit-Circle, Bessel, and Spectral–Monodromy Representations

A comprehensive atlas for the Rvachev and Fabius functions, their Fourier images, and the Thue–Morse sequence

Research report

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Abstract

This report audits the mathematical L^AT_EX corpus under <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>, including its subdirectories, source inventories, historical snapshots, imported papers, and living frontier reports. It first consolidates the repository’s established probabilistic, convolutional, spline, Fourier, product, moment, resolvent, continued-fraction, multiresolution, integral-transform, and inverse-function representations. It then develops a repo-relative new coordinate system based on the probability–Chebyshev moments

$$c_n = \int_{-1}^1 T_n(x) \operatorname{up}(x) \, dx.$$

These moments lift the Rvachev probability measure to a symmetric probability measure on the unit circle. The lift yields an exact Carathéodory function, a Joukowski representation of the Cauchy transform, weighted Chebyshev boundary series for up , phase series for the Fabius function and its inverse, rational Schur–Verblunsky parameters, a CMV resolvent, Toeplitz determinants, and a Szegő product.

Jacobi–Anger expansions produce entire Bessel-series identities equal to the infinite sinc product and its canonical dyadic zero product, while modified Bessel series equal the hyperbolic moment-generating product. A Laplace–Bessel–Joukowski chain unifies these identities with the Cauchy transform. For the signed and unsigned Thue–Morse sequences, we derive argument-principle formulas from the multiplicities of the integer zeros of the Fourier image, local residue recurrences, a logarithmic-monodromy recovery formula for finite blocks, and a smooth Abel–Poisson regularization of the Rademacher sign product.

Every theorem marked “repo-relative new” is proved here and was not located by exact-term or structural search in the audited repository. This phrase is deliberately narrower than a claim of worldwide priority. Conjectures and a research program are stated separately. The accompanying Python program uses exact rational arithmetic and high-precision numerics to verify the coefficient pipelines and analytic identities.

Status and reproducibility

The archive includes the full L^AT_EX source, exact coefficient CSVs, generated tables, three convergence/parameter figures, a commented Python program, a numerical verification log, and a source-overlap ledger. The direct Schur algorithm and the independent Geronimus reconstruction agree exactly through the checked orders; the spectral Thue–Morse formulas

pass for $0 \leq n \leq 256$; and three independent Cauchy-transform evaluations agree to the precision recorded in Section 9.

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1 Scope, conventions, and nonduplication protocol

1.1 Normalized objects

We use the Rvachev probability density

$$\rho(x) := \text{up}(x), \quad \rho \geq 0, \quad \text{supp } \rho = [-1, 1], \quad \int_{-1}^1 \rho(x) \, dx = 1,$$

which is even and C^∞ , with every derivative flat at the endpoints. Its associated random variable admits the convergent series

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k, \quad U_k \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1]. \quad (1)$$

Set

$$Y = \frac{X+1}{2}, \quad F(u) = \mathbb{P}(Y \leq u) = \int_{-1}^{2u-1} \rho(x) \, dx, \quad I(v) = F^{-1}(v).$$

Thus, on $0 < u < 1$,

$$F'(u) = 2\rho(2u-1). \quad (2)$$

The Fourier convention is

$$\Phi(\xi) = \int_{-1}^1 \rho(x) e^{-2\pi i \xi x} \, dx, \quad \text{sinc}_\pi z = \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}_\pi 0 = 1. \quad (3)$$

The moment-generating and Cauchy transforms are

$$M(t) = \mathbb{E} e^{tX}, \quad R(z) = \int_{-1}^1 \frac{\rho(x)}{z-x} \, dx, \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (4)$$

For the Thue–Morse sequence we use

$$\epsilon_n = (-1)^{s_2(n)}, \quad \tau_n = \frac{1 - \epsilon_n}{2} \in \{0, 1\}, \quad (5)$$

where $s_2(n)$ is the sum of the binary digits of n .

1.2 Corpus audit

The current repository manifest organizes the documentation into Fourier-decay, Thue–Morse, exponent and q -series, spectral/arithmetic, integration and transform, inverse/sampling, representation, and frontier campaigns. The repository’s own pinned corpus inventory enumerates sixty-seven source documents plus later living additions, while the multiresolution report records a source-level audit of eighty-nine L^AT_EX paths at commit 924652f4a9a656fd957d2936b2f053747c6f6c47. The consolidated frontier volume, the current manifest, and the overlap-sensitive living reports were used jointly so that historical duplicates did not masquerade as independent novelty.

The following families are already present and are therefore treated as baseline rather than novelty: the infinite sinc product; the canonical zero product; finite box-convolution splines; finite Thue–Morse truncated-power formulas; Cauchy and logarithmic resolvents; moment/cumulant and Bell–Bernoulli

formulas; transforms of F and F^{-1} ; digamma, Hurwitz-zeta, and cotangent logarithmic derivatives; Fredholm determinants; Jacobi and Stieltjes continued fractions; Haar, Schauder, Walsh, and beta-mixture expansions; fractional Volterra integrals; Mellin/Newton moments; inverse endpoint asymptotics; the half-integer sign formula; dyadic cardinal reproduction; and the non-holonomicity theorem from unbounded dyadic zero multiplicities.

The exact terms and structures *Carathéodory function*, *OPUC*, *Verblunsky coefficient*, *CMV matrix*, *Joukowski transform*, *probability–Chebyshev moment*, *Bessel product-series*, *argument-principle Thue–Morse formula*, and *logarithmic-monodromy encoding* were not found in the current consolidated volume or in the relevant living reports. In particular, the multiresolution report’s density-Chebyshev coefficients are different from the probability-Chebyshev moments introduced in Section 3.

Repository baseline

The audit is documented in `corpus_audit.txt`. Throughout the report, “new” means “not located in the audited PROVEIT subtree.” Standard external machinery—Jacobi–Anger expansions, the Schur algorithm, Geronimus relations, Szegő’s theorem, CMV matrices, Shannon sampling, and the argument principle—is cited rather than claimed as new.

1.3 Proof-status convention

A theorem or proposition without qualification is proved in this report. A boxed “repository baseline” records known material used as input. Conjectures are numbered and visually separated. Numerical evidence is never promoted to a proof; exact rational calculations are identified as such.

2 Established representation atlas

This section compresses the repository’s existing representation families into one normalization. It is included both for completeness and to make the genuinely new bridges in later sections transparent.

2.1 Random series, convolutions, and refinement

From (1), X satisfies the distributional fixed point

$$X \stackrel{d}{=} \frac{X + U}{2}, \quad U \sim \text{Unif}[-1, 1], \quad U \perp X. \quad (6)$$

Consequently ρ is an infinite convolution of normalized boxes,

$$\rho = \bigast_{k \geq 1} \left(2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]} \right), \quad (7)$$

and the finite truncation $X_N = \sum_{k=1}^N 2^{-k} U_k$ has a compactly supported polynomial-spline density ρ_N . The refinement and differential identities can be written

$$\rho(x) = \int_{-1}^1 \rho(2x - u) du, \quad (8)$$

$$\rho'(x) = 2(\rho(2x + 1) - \rho(2x - 1)). \quad (9)$$

They imply the familiar Fabius functional-differential relations after the affine change $x = 2u - 1$.

2.2 Finite Thue–Morse splines

Let

$$a_{N,j} = \frac{2^N - 1 - 2j}{2^N}, \quad 0 \leq j < 2^N.$$

The finite convolution density has the exact truncated-power form

$$\rho_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \epsilon_j (x + a_{N,j})_+^{N-1}. \quad (10)$$

The cancellation of all polynomial pieces below degree N is a Prouhet–Thue–Morse phenomenon. Equivalently,

$$P_N(z) := \prod_{r=0}^{N-1} (1 - z^{2^r}) = \sum_{j=0}^{2^N-1} \epsilon_j z^j. \quad (11)$$

The corresponding unsigned block has

$$\sum_{j=0}^{2^N-1} \tau_j z^j = \frac{1}{2} \left(\frac{1 - z^{2^N}}{1 - z} - P_N(z) \right). \quad (12)$$

2.3 Fourier, hyperbolic, and canonical products

Independence in (1) gives

$$\Phi(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_{\pi} \left(\frac{\xi}{2^j} \right), \quad (13)$$

$$M(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k}. \quad (14)$$

Rearranging Euler products yields

$$\Phi(\xi) = \prod_{m=1}^{\infty} \left(1 - \frac{\xi^2}{m^2} \right)^{1+\nu_2(m)}, \quad (15)$$

$$M(t) = \prod_{m=1}^{\infty} \left(1 + \frac{t^2}{4\pi^2 m^2} \right)^{1+\nu_2(m)}. \quad (16)$$

Thus Φ has a zero at every nonzero integer m , of multiplicity

$$d_m = 1 + \nu_2(|m|). \quad (17)$$

Its logarithmic derivative is the normally convergent meromorphic series

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^2 - z^2}. \quad (18)$$

The audited repository develops several equivalent cotangent, tangent, digamma, and Hurwitz-zeta rearrangements of this expression.

2.4 Moments, cumulants, Bell polynomials, and Bernoulli numbers

Write $\mu_n = \mathbb{E}X^n$. Symmetry gives $\mu_{2m+1} = 0$. Applying (6) gives the exact triangular recurrence

$$\mu_n = \frac{1}{2^n - 1} \sum_{\substack{2 \leq k \leq n \\ k \text{ even}}} \binom{n}{k} \frac{\mu_{n-k}}{k+1} \quad (n \geq 2 \text{ even}). \quad (19)$$

The cumulants are

$$\kappa_{2m} = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad \kappa_{2m+1} = 0, \quad (20)$$

and therefore

$$\mu_n = Y_n(\kappa_1, \dots, \kappa_n), \quad (21)$$

where Y_n is the complete exponential Bell polynomial. These formulas make all moments rational and expose the Bernoulli and dyadic denominator structure.

2.5 Cauchy, logarithmic, determinant, and continued-fraction forms

For $z \notin [-1, 1]$, the Cauchy transform has the moment expansion

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1, \quad (22)$$

and, for $\Re z > 1$, the Laplace form

$$R(z) = \int_0^{\infty} e^{-zt} M(t) dt. \quad (23)$$

Differentiating under the integral or using (9) gives

$$R'(z) = 2(R(2z+1) - R(2z-1)). \quad (24)$$

The finite spline transform possesses the logarithmic formula

$$R_N(z) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \epsilon_j (z + a_{N,j})^{N-1} \text{Log}(z + a_{N,j}), \quad (25)$$

with polynomial ambiguities canceled by Prouhet identities. The living representation report also gives Fredholm determinant and Jacobi/Stieltjes continued-fraction forms. If P_n are the monic symmetric orthogonal polynomials,

$$P_{n+1}(x) = xP_n(x) - \beta_n P_{n-1}(x), \quad \beta_n > 0, \quad (26)$$

then

$$R(z) = \frac{1}{z - \frac{\beta_1}{z - \frac{\beta_2}{z - \ddots}}}. \quad (27)$$

2.6 Fabius and inverse-function transform forms

The pushforward identity $Y = (X + 1)/2$ yields

$$\int_0^1 e^{-2\pi i \xi u} dF(u) = e^{-\pi i \xi} \Phi(\xi/2). \quad (28)$$

For $\xi \neq 0$, integration by parts gives

$$\int_0^1 F(u) e^{-2\pi i \xi u} du = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (29)$$

Since $I(V)$ has distribution F when $V \sim \text{Unif}[0, 1]$,

$$\int_0^1 e^{-2\pi i \xi I(v)} dv = e^{-\pi i \xi} \Phi(\xi/2). \quad (30)$$

The repository further contains finite dyadic antiderivatives, complex powers, fractional Volterra operators, Mellin/Newton moments, inverse primitives and higher derivatives, Stieltjes/log-potential transforms, autocorrelation identities, and endpoint asymptotics.

Repository baseline

Equations (7)–(30) form the input atlas. Later sections do not claim these identities as new; they reorganize them through a unit-circle probability lift and derive consequences absent from the audited subtree.

3 Probability–Chebyshev coordinates

Definition 3.1. For $n \geq 0$, define the *probability–Chebyshev moment*

$$c_n := \mathbb{E}[T_n(X)] = \int_{-1}^1 T_n(x) \rho(x) dx, \quad (31)$$

where $T_n(\cos \theta) = \cos(n\theta)$.

Because ρ is even, $c_{2m+1} = 0$. These numbers are not the unweighted Chebyshev coefficients of the density: the integration measure in (31) is $\rho(x) dx$, not $dx/\sqrt{1-x^2}$.

Theorem 3.2 (Exact Bell–Bernoulli formula). *For $m \geq 1$,*

$$c_{2m} = m \sum_{k=0}^m (-1)^{m-k} \frac{(m+k-1)!}{(m-k)!(2k)!} 2^{2k} \mu_{2k}. \quad (32)$$

Equivalently, with κ_{2j} as in (20),

$$c_{2m} = m \sum_{k=0}^m (-1)^{m-k} \frac{(m+k-1)!}{(m-k)!(2k)!} 2^{2k} Y_{2k}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2k}). \quad (33)$$

In particular every c_n is rational.

Proof. For $m \geq 1$,

$$T_{2m}(x) = m \sum_{k=0}^m (-1)^{m-k} \frac{(m+k-1)!}{(m-k)!(2k)!} (2x)^{2k}.$$

Integrating termwise against $\rho(x) dx$ proves (32); substituting (21) and (20) proves (33). $\square$

The first terms are

$$\begin{aligned} c_0 &= 1, & c_2 &= -\frac{7}{9}, & c_4 &= \frac{227}{675}, & c_6 &= -\frac{11237}{297675}, \\ c_8 &= -\frac{8241761}{227721375}, & c_{10} &= \frac{171586753}{8135075025}, & c_{12} &= -\frac{472381023339557}{45472425490366875}. \end{aligned}$$

The repeated negative sign at c_6, c_8 warns against a naive alternating-sign conjecture.

Proposition 3.3 (Generating function). *For $|w| < 1$,*

$$G(w) := \sum_{n=0}^{\infty} c_n w^n = \int_{-1}^1 \frac{1 - wx}{1 - 2wx + w^2} \rho(x) dx, \quad (34)$$

$$= 1 + \sum_{m=1}^{\infty} c_{2m} w^{2m}. \quad (35)$$

Proof. Integrate the standard generating function $\sum_{n \geq 0} T_n(x) w^n = (1 - wx)/(1 - 2wx + w^2)$ against the probability measure. $\square$

For any $0 < r < 1$, Cauchy coefficient extraction gives

$$c_n = \frac{1}{2\pi r^n} \int_0^{2\pi} G(re^{i\theta}) e^{-in\theta} d\theta. \quad (36)$$

This will become an ordinary circle Fourier coefficient in the next section.

Repo-relative new result

The sequence c_n is one rational coordinate system controlling a positive-real Carathéodory function, the Joukowski pullback of R , a weighted Chebyshev expansion of ρ , phase series for F and F^{-1} , Bessel expansions of Φ and M , and rational OPUC/CMV data. This unification is the central structural result of the report.

4 Unit-circle lift and Carathéodory–Joukowski bridge

4.1 A canonical probability measure on the circle

Define a probability measure ν on $\mathbb{T}$ by

$$d\nu(\theta) = \frac{1}{2} \rho(\cos \theta) |\sin \theta| d\theta, \quad 0 \leq \theta < 2\pi. \quad (37)$$

The map $\theta \mapsto x = \cos \theta$ pushes ν forward to $\rho(x) dx$. Moreover, ν is invariant under conjugation and the half-turn $\zeta \mapsto -\zeta$.

Proposition 4.1 (Fourier moments of the lift). *For every $n \in \mathbb{Z}$,*

$$\int_{\mathbb{T}} \zeta^{-n} d\nu(\zeta) = c_{|n|}. \quad (38)$$

Proof. Imaginary parts cancel under conjugation, while $\cos(n \arccos x) = T_n(x)$. The two preimages of each $x \in (-1, 1)$ contribute equally. $\square$

Definition 4.2. The Carathéodory function of ν is

$$C(w) = \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} d\nu(\zeta), \quad w \in \mathbb{D}. \quad (39)$$

It is analytic in $\mathbb{D}$, satisfies $C(0) = 1$, and has positive real part.

Theorem 4.3 (Carathéodory representation). *For $|w| < 1$,*

$$C(w) = \int_{-1}^1 \frac{1 - w^2}{1 - 2wx + w^2} \rho(x) dx, \quad (40)$$

$$= 1 + 2 \sum_{n=1}^{\infty} c_n w^n = 1 + 2 \sum_{m=1}^{\infty} c_{2m} w^{2m}, \quad (41)$$

$$= 2G(w) - 1. \quad (42)$$

Proof. Pair $e^{i\theta}$ and $e^{-i\theta}$ in (39); their average kernel is $(1 - w^2)/(1 - 2w \cos \theta + w^2)$. The geometric expansion $(\zeta + w)/(\zeta - w) = 1 + 2 \sum_{n \geq 1} w^n \zeta^{-n}$ and Theorem 4.1 give the series. $\square$

The positive harmonic regularization

$$W_r(\theta) := \Re C(re^{i\theta}) = 1 + 2 \sum_{m=1}^{\infty} c_{2m} r^{2m} \cos(2m\theta) \quad (43)$$

preserves positivity for every $0 \leq r < 1$, unlike an arbitrary boundary truncation.

4.2 Joukowski uniformization

Let

$$\mathcal{J}(w) = \frac{1}{2}(w + w^{-1}), \quad 0 < |w| < 1, \quad (44)$$

with inverse branch

$$w(z) = z - \sqrt{z^2 - 1}, \quad \sqrt{z^2 - 1} \sim z \quad (z \rightarrow \infty). \quad (45)$$

Theorem 4.4 (Exact Joukowski bridge). *For $0 < |w| < 1$,*

$$C(w) = \frac{1 - w^2}{2w} R(\mathcal{J}(w)). \quad (46)$$

Equivalently, for $z \in \mathbb{C} \setminus [-1, 1]$,

$$R(z) = \frac{2w(z)}{1 - w(z)^2} C(w(z)), \quad (47)$$

$$= \frac{1 + 2 \sum_{m=1}^{\infty} c_{2m} w(z)^{2m}}{\sqrt{z^2 - 1}}. \quad (48)$$

Proof. Since $\mathcal{J}(w) - x = (1 - 2wx + w^2)/(2w)$, compare (40) with the definition of R . On the chosen branch, $\sqrt{z^2 - 1} = (1 - w^2)/(2w)$. $\square$

The ordinary moment series is centered at infinity in powers of z^{-1} . By contrast, (48) uses the conformal coordinate of the slit plane, so it remains geometrically adapted even near the support.

4.3 Boundary recovery of ρ

Set

$$W(\theta) = \pi\rho(\cos\theta)|\sin\theta|. \quad (49)$$

This is the density of ν relative to $d\theta/(2\pi)$. Endpoint flatness makes it a C^∞ , π -periodic function.

Theorem 4.5 (Weighted Chebyshev boundary series). *For every θ ,*

$$W(\theta) = \Re C(e^{i\theta}) = 1 + 2 \sum_{m=1}^{\infty} c_{2m} \cos(2m\theta), \quad (50)$$

with convergence in C^k for every fixed k . Consequently, for $|x| < 1$,

$$\boxed{\rho(x) = \frac{1 + 2 \sum_{m=1}^{\infty} c_{2m} T_{2m}(x)}{\pi \sqrt{1 - x^2}}.} \quad (51)$$

Proof. The boundary theorem for Poisson integrals gives $\Re C(e^{i\theta}) = W(\theta)$. Smoothness implies uniform convergence of all fixed derivatives. Substitute $x = \cos\theta$. $\square$

The harmonic conjugate is

$$\Im C(e^{i\theta}) = 2 \sum_{m=1}^{\infty} c_{2m} \sin(2m\theta), \quad (52)$$

$$= \text{PV} \int_0^{2\pi} \cot\left(\frac{\theta - \varphi}{2}\right) d\nu(\varphi). \quad (53)$$

Thus one disk-analytic function packages the density and its circular Hilbert transform.

4.4 Phase series for F and F^{-1}

For $0 < u < 1$, put

$$\theta(u) = \arccos(2u - 1) \in (0, \pi). \quad (54)$$

Theorem 4.6 (Fabius phase series). *For $0 \leq u \leq 1$,*

$$\boxed{F(u) = 1 - \frac{\theta(u)}{\pi} - \frac{1}{\pi} \sum_{m=1}^{\infty} \frac{c_{2m}}{m} \sin(2m\theta(u)).} \quad (55)$$

For $0 < u < 1$,

$$F'(u) = \frac{\Re C(e^{i\theta(u)})}{\pi \sqrt{u(1-u)}} = \frac{1 + 2 \sum_{m \geq 1} c_{2m} T_{2m}(2u - 1)}{\pi \sqrt{u(1-u)}}. \quad (56)$$

Proof. Substitute $x = \cos \varphi$ in $F(u) = \int_{-1}^{2u-1} \rho(x) dx$: $F(u) = \pi^{-1} \int_{\theta(u)}^{\pi} W(\varphi) d\varphi$. Integrate (50) termwise and differentiate. $\square$

Corollary 4.7 (Inverse phase equation). *For each $v \in (0, 1)$, let $\Theta(v) \in (0, \pi)$ be the unique solution of*

$$v = 1 - \frac{\Theta}{\pi} - \frac{1}{\pi} \sum_{m=1}^{\infty} \frac{c_{2m}}{m} \sin(2m\Theta). \quad (57)$$

Then

$$I(v) = \frac{1 + \cos \Theta(v)}{2}, \quad (58)$$

$$I'(v) = \frac{\pi \sqrt{I(v)(1 - I(v))}}{\Re C(e^{i\Theta(v)})}. \quad (59)$$

The explicit factors $1/\sqrt{u(1-u)}$ do not create endpoint singularities in F' : the numerator W is flat enough to cancel every algebraic divergence. In the inverse formula, $1/W$ isolates the true rapid endpoint growth and is compatible with the Lambert– W asymptotics in the repository.

4.5 Disk form of the resolvent differential equation

Set

$$A(w) = \frac{2wC(w)}{1 - w^2} = R(\mathcal{J}(w)) \quad (60)$$

and, as germs at $w = 0$,

$$q_{\pm}(w) = 2\mathcal{J}(w) \pm 1 - \sqrt{(2\mathcal{J}(w) \pm 1)^2 - 1}, \quad (61)$$

choosing square roots so that $q_{\pm}(w) \rightarrow 0$.

Proposition 4.8 (Disk composition equation). *Near $w = 0$,*

$$\frac{d}{dw} \left(\frac{2wC(w)}{1 - w^2} \right) = 4\mathcal{J}'(w) \left[\frac{q_+(w)C(q_+(w))}{1 - q_+(w)^2} - \frac{q_-(w)C(q_-(w))}{1 - q_-(w)^2} \right]. \quad (62)$$

Proof. Differentiate $A(w) = R(\mathcal{J}(w))$, apply (24), and use $R(\mathcal{J}(q)) = 2qC(q)/(1 - q^2)$ for each inverse branch. $\square$

5 Bessel series equal to the dyadic products

The Jacobi–Anger expansions

$$e^{-iz \cos \theta} = J_0(z) + 2 \sum_{n=1}^{\infty} (-i)^n J_n(z) \cos(n\theta), \quad (63)$$

$$e^{z \cos \theta} = I_0(z) + 2 \sum_{n=1}^{\infty} I_n(z) \cos(n\theta) \quad (64)$$

turn the Fourier moments of ν into Bessel coefficients.

Theorem 5.1 (Bessel–sinc–canonical product identity). *For every $\xi \in \mathbb{C}$,*

$$\Phi(\xi) = J_0(2\pi\xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(2\pi\xi). \quad (65)$$

The series is locally uniformly convergent, and

$$J_0(2\pi\xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(2\pi\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{\xi}{2^j}\right) \quad (66)$$

$$= \prod_{n=1}^{\infty} \left(1 - \frac{\xi^2}{n^2}\right)^{1+\nu_2(n)}. \quad (67)$$

Proof. Under ν , $X = \cos \Theta$. Integrate (63) with $z = 2\pi\xi$; odd moments vanish and $(-i)^{2m} = (-1)^m$. The product equalities are (13) and (15). $\square$

For every $r \geq 0$, termwise differentiation gives

$$\Phi^{(r)}(\xi) = (2\pi)^r \left[J_0^{(r)}(2\pi\xi) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}^{(r)}(2\pi\xi) \right]. \quad (68)$$

At an integer zero $N \neq 0$, the first $1 + \nu_2(N)$ equations in this derivative ladder become exact Bessel cancellation identities. In particular,

$$J_0(2\pi N) + 2 \sum_{m=1}^{\infty} (-1)^m c_{2m} J_{2m}(2\pi N) = 0. \quad (69)$$

Theorem 5.2 (Modified-Bessel–hyperbolic identity). *For every $t \in \mathbb{C}$,*

$$M(t) = I_0(t) + 2 \sum_{m=1}^{\infty} c_{2m} I_{2m}(t). \quad (70)$$

Moreover,

$$I_0(t) + 2 \sum_{m=1}^{\infty} c_{2m} I_{2m}(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} \quad (71)$$

$$= \prod_{n=1}^{\infty} \left(1 + \frac{t^2}{4\pi^2 n^2}\right)^{1+\nu_2(n)}. \quad (72)$$

Proof. Integrate (64) against ν and invoke the product formulas for M . $\square$

5.1 Laplace–Bessel–Joukowski chain

For $\Re z > 1$,

$$\int_0^{\infty} e^{-zt} I_n(t) dt = \frac{(z - \sqrt{z^2 - 1})^n}{\sqrt{z^2 - 1}}. \quad (73)$$

Theorem 5.3 (Three-level transform identity). *For $\Re z > 1$, with $w = z - \sqrt{z^2 - 1}$,*

$$R(z) = \int_0^\infty e^{-zt} \prod_{k=1}^\infty \frac{\sinh(t/2^k)}{t/2^k} dt \quad (74)$$

$$= \int_0^\infty e^{-zt} \left[I_0(t) + 2 \sum_{m=1}^\infty c_{2m} I_{2m}(t) \right] dt \quad (75)$$

$$= \frac{1 + 2 \sum_{m=1}^\infty c_{2m} w^{2m}}{\sqrt{z^2 - 1}}. \quad (76)$$

Proof. Insert (14) and (70) into (23); use dominated convergence and (73). $\square$

5.2 Fabius and inverse-Fabius transforms

For $\xi \neq 0$,

$$\int_0^1 F(u) e^{-2\pi i \xi u} du = \frac{e^{-\pi i \xi} [J_0(\pi \xi) + 2 \sum_{m=1}^\infty (-1)^m c_{2m} J_{2m}(\pi \xi)] - e^{-2\pi i \xi}}{2\pi i \xi}. \quad (77)$$

The value at $\xi = 0$ is $1/2$. For the inverse pushforward,

$$\int_0^1 e^{-2\pi i \xi I(v)} dv = e^{-\pi i \xi} \left[J_0(\pi \xi) + 2 \sum_{m=1}^\infty (-1)^m c_{2m} J_{2m}(\pi \xi) \right]. \quad (78)$$

Repo-relative new result

Equations (66), (67), (71), and (72) are direct series–product bridges: the same rational sequence c_{2m} expands both the Fourier image and the moment generating function in Bessel bases. No corresponding bridge was located in the audited repository.

6 Schur, Verblunsky, CMV, and Toeplitz representations

The Carathéodory function places the Rvachev measure in the theory of orthogonal polynomials on the unit circle (OPUC) [5, 6]. This creates a rational coefficient system distinct from, but explicitly coupled to, the repository’s Jacobi coefficients.

6.1 Schur algorithm and rationality

Define

$$f_0(w) = \frac{C(w) - 1}{w(C(w) + 1)}, \quad C(w) = \frac{1 + w f_0(w)}{1 - w f_0(w)}. \quad (79)$$

Recursively set

$$\alpha_n = f_n(0), \quad f_{n+1}(w) = \frac{f_n(w) - \alpha_n}{w(1 - \alpha_n f_n(w))}. \quad (80)$$

All functions and coefficients are real under the present symmetries.

Theorem 6.1 (Rational Verblunsky coefficients). *The Schur parameters α_n are the Verblunsky coefficients of ν . They satisfy*

$$\alpha_n \in \mathbb{Q} \cap (-1, 1), \quad \alpha_{2n} = 0 \quad (n \geq 0). \quad (81)$$

The odd coefficients are obtained by exact rational operations from the moments c_{2m} .

Proof. The first assertion is the standard Carathéodory–Schur correspondence. Rationality follows because the Taylor coefficients of C are rational and every Schur step uses rational operations. The half-turn invariance $\nu(E) = \nu(-E)$ forces all odd circle moments and, equivalently, every even-index Verblunsky coefficient to vanish. Infinite support gives $|\alpha_n| < 1$. $\square$

The first nonzero values are

$$\begin{aligned} \alpha_1 &= -\frac{7}{9}, & \alpha_3 &= -\frac{17}{25}, & \alpha_5 &= -\frac{1849}{3087}, \\ \alpha_7 &= -\frac{3569379}{6492691}, & \alpha_9 &= -\frac{445076883510763}{886612917691413}, & \alpha_{11} &= -\frac{594580974953911037169143}{1270428317522023221443375}. \end{aligned}$$

The exact program computes much longer coefficients without floating-point contamination.

6.2 Wall–Schur continued fraction

Solving (80) for f_n gives

$$f_n(w) = \frac{\alpha_n + w f_{n+1}(w)}{1 + \alpha_n w f_{n+1}(w)}. \quad (82)$$

Iterating this Möbius recursion and inserting it into (79) produces a Wall–Schur continued fraction for C . Truncating after N parameters by setting $f_N = 0$ gives a rational C_N with positive real part in $\mathbb{D}$. Therefore

$$R_N^{\text{Schur}}(z) = \frac{C_N(w(z))}{\sqrt{z^2 - 1}} \quad (83)$$

is a positivity-preserving rational/conformal approximation to the Cauchy transform. It is geometrically different from truncating the real-line Jacobi fraction (27).

6.3 Geronimus bridge to Jacobi coefficients

For the symmetric pushforward under $x = (\zeta + \zeta^{-1})/2$, the Geronimus relation simplifies because $\alpha_{2n} = 0$.

Theorem 6.2 (Exact Jacobi–Verblunsky conversion). *With $\alpha_{-1} = -1$, the monic Jacobi coefficients satisfy*

$$\boxed{\beta_{n+1} = \frac{1}{4}(1 - \alpha_{2n-1})(1 + \alpha_{2n+1})} \quad (n \geq 0). \quad (84)$$

Equivalently,

$$\alpha_{2n+1} = \frac{4\beta_{n+1}}{1 - \alpha_{2n-1}} - 1. \quad (85)$$

The first values

$$\beta_1 = \frac{1}{9}, \quad \beta_2 = \frac{32}{225}, \quad \beta_3 = \frac{619}{3675}, \quad \beta_4 = \frac{835232}{4640643} \quad (86)$$

reproduce $\alpha_1, \alpha_3, \alpha_5, \alpha_7$. In the accompanying code, the β_n are generated by exact Gram–Schmidt from μ_n , while a direct formal Schur algorithm independently generates α_n ; the pipelines agree exactly through the checked indices.

6.4 CMV resolvent

Let $\mathcal{C}$ be the CMV matrix associated with $(\alpha_n)_{n \geq 0}$ on $\ell^2(\mathbb{N}_0)$, and let $e_0 = (1, 0, \dots)$. Its spectral measure at e_0 is ν .

Theorem 6.3 (CMV representation). *For $|w| < 1$,*

$$C(w) = \left\langle e_0, (\mathcal{C} + w)(\mathcal{C} - w)^{-1} e_0 \right\rangle. \quad (87)$$

Consequently,

$$R(\mathcal{J}(w)) = \frac{2w}{1 - w^2} \left\langle e_0, (\mathcal{C} + w)(\mathcal{C} - w)^{-1} e_0 \right\rangle. \quad (88)$$

Proof. Apply the spectral theorem to multiplication by ζ in $L^2(\nu)$, then express that unitary operator in the CMV basis. The second formula follows from (46). $\square$

6.5 Toeplitz determinants and Szegő product

Let

$$D_N = \det[c_{j-k}]_{j,k=0}^{N-1}, \quad c_{-n} = c_n. \quad (89)$$

These determinants are positive rationals.

Proposition 6.4 (Exact determinant product). *For $N \geq 1$,*

$$D_N = \prod_{j=0}^{N-2} (1 - \alpha_j^2)^{N-1-j}. \quad (90)$$

Only odd j contribute.

The weight $W(\theta)$ is positive almost everywhere and $\log W \in L^1(\mathbb{T})$. The latter follows from the established endpoint asymptotics: although $\rho(\cos \theta)$ is superflat, its logarithm grows only polylogarithmically in $1/|\theta|$, which is integrable against $d\theta$. Szegő’s theorem gives

$$\prod_{n=0}^{\infty} (1 - \alpha_n^2) = \exp \left(\frac{1}{2\pi} \int_0^{2\pi} \log W(\theta) d\theta \right). \quad (91)$$

Rakhmanov’s theorem also yields $\alpha_n \rightarrow 0$, because $W > 0$ almost everywhere.

Repo-relative new result

Equations (87), (88), (90), and (91) give operator, determinant, and entropy-product representations of the Rvachev law. The associated CMV coefficients are exact rationals and are linked to the known Jacobi coefficients by (84).

7 Spectral, contour, and monodromy formulas for Thue–Morse

This section derives representations for both ϵ_n and τ_n by treating the integer zeros of Φ as spectral data. These formulas differ from the repository’s coefficient, digit, half-integer-sign, trigonometric, and finite-product formulas.

7.1 Zero-counting encodes the full binary digit sum

Let

$$N_\Phi(n) := \sum_{m=1}^n (1 + \nu_2(m)), \quad N_\Phi(0) = 0. \quad (92)$$

This is the number of positive zeros of Φ in $(0, n + 1/2)$, counted with multiplicity.

Theorem 7.1 (Spectral digit-sum identity). *For every $n \geq 0$,*

$$N_\Phi(n) = n + \nu_2(n!) = 2n - s_2(n). \quad (93)$$

Thus the complete binary digit sum, not only its parity, is recovered by

$$\boxed{s_2(n) = 2n - N_\Phi(n)}. \quad (94)$$

Proof. The first equality follows from (17). Legendre’s formula for the 2-adic valuation gives $\nu_2(n!) = n - s_2(n)$. $\square$

Let Γ_n be any positively oriented contour enclosing precisely the positive integer zeros $1, \dots, n$, and no other zeros of Φ .

Corollary 7.2 (Argument-principle representations). *For $n \geq 0$,*

$$s_2(n) = 2n - \frac{1}{2\pi i} \oint_{\Gamma_n} \frac{\Phi'(z)}{\Phi(z)} dz, \quad (95)$$

$$\boxed{\epsilon_n = \exp \left(\frac{1}{2} \oint_{\Gamma_n} \frac{\Phi'(z)}{\Phi(z)} dz \right)}, \quad (96)$$

$$\boxed{\tau_n = \sin^2 \left(\frac{1}{4i} \oint_{\Gamma_n} \frac{\Phi'(z)}{\Phi(z)} dz \right)}. \quad (97)$$

Proof. The argument principle gives the contour integral as $2\pi i N_\Phi(n)$. Since $N_\Phi(n) \equiv s_2(n) \pmod{2}$, exponentiation gives ϵ_n , while $\sin^2(\pi N_\Phi(n)/2)$ is its unsigned parity. $\square$

Substitution of (18) turns the contour formulas into explicit meromorphic series integrals. Locally at a positive integer,

$$\operatorname{Res}_{z=n} \frac{\Phi'(z)}{\Phi(z)} = 1 + \nu_2(n), \quad (98)$$

and hence

$$\frac{\epsilon_n}{\epsilon_{n-1}} = \exp \left(\pi i \operatorname{Res}_{z=n} \frac{\Phi'(z)}{\Phi(z)} \right) = (-1)^{1+\nu_2(n)}. \quad (99)$$

Thus the local zero multiplicity records the carry chain when passing from $n - 1$ to n .

7.2 Logarithmic monodromy of finite resolvents

The finite resolvent (25) has branch points

$$z_{N,j} = -a_{N,j}, \quad 0 \leq j < 2^N. \quad (100)$$

Let $\Delta_j R_N$ denote the change after one positive analytic continuation around $z_{N,j}$, holding all other logarithmic branches fixed.

Theorem 7.3 (Monodromy recovery of a finite Thue–Morse word). *For z near $z_{N,j}$,*

$$\Delta_j R_N(z) = 2\pi i \frac{2^{N(N-1)/2}}{(N-1)!} \epsilon_j (z - z_{N,j})^{N-1}. \quad (101)$$

Consequently,

$$\epsilon_j = \frac{1}{2\pi i 2^{N(N-1)/2}} \left. \frac{d^{N-1}}{dz^{N-1}} \Delta_j R_N(z) \right|_{z=z_{N,j}}, \quad (102)$$

and $\tau_j = (1 - \epsilon_j)/2$.

Proof. Only the j -th logarithm changes, by $2\pi i$, around $z_{N,j}$. This proves (101). Differentiating $N - 1$ times at the branch point cancels $(N - 1)!$ and isolates ϵ_j . $\square$

The finite Thue–Morse word is therefore the normalized monodromy representation of a single multivalued analytic function. This interpretation is stronger than saying that ϵ_j merely occurs as an algebraic coefficient: the word is recovered from the analytic continuation operators themselves.

7.3 Abel–Poisson regularization of the Rademacher product

Define, for $0 < r < 1$,

$$H_r(\theta) := \frac{4}{\pi} \sum_{\ell=0}^{\infty} \frac{r^{2\ell+1}}{2\ell+1} \sin((2\ell+1)\theta) \quad (103)$$

$$= \frac{2}{\pi} \operatorname{atan2}(2r \sin \theta, 1 - r^2). \quad (104)$$

This is the Abel/Poisson regularization of $\operatorname{sgn}(\sin \theta)$, and $H_r(\theta) \rightarrow \operatorname{sgn}(\sin \theta)$ away from multiples of π .

For $n \geq 1$, put $J_n = \lfloor \log_2(n + 1/2) \rfloor$; for $n = 0$, interpret the product below as empty.

Theorem 7.4 (Smooth finite-product representation). *For every $n \geq 0$,*

$$\epsilon_n = \lim_{r \uparrow 1} \prod_{j=0}^{J_n} H_r \left(\frac{\pi(n + 1/2)}{2^j} \right), \quad (105)$$

$$\tau_n = \frac{1}{2} \left[1 - \lim_{r \uparrow 1} \prod_{j=0}^{J_n} H_r \left(\frac{\pi(n + 1/2)}{2^j} \right) \right]. \quad (106)$$

Proof. The signed Thue–Morse sequence has the Rademacher sign product

$$\epsilon_n = \prod_{j \geq 0} \operatorname{sgn} \sin \left(\frac{\pi(n + 1/2)}{2^j} \right).$$

For $j > J_n$, the angle lies in $(0, \pi)$, so the factor is $+1$; only finitely many factors matter. Replace each by its Abel limit. $\square$

The accompanying computation shows uniform convergence over the first 64 symbols: the maximum errors for $r = 0.9, 0.99, 0.999, 0.9999$ are approximately 0.937, 0.237, 0.0257, 0.00259, respectively. The formula is especially useful when a smooth analytic surrogate is preferable to a discontinuous sign function.

7.4 A cardinal Bessel–Thue–Morse expansion

Since ρ is supported in $[-1, 1]$, the Paley–Wiener/Shannon formula at spacing $1/2$ is

$$\Phi(z) = \sum_{k \in \mathbb{Z}} \Phi(k/2) \operatorname{sinc}_\pi(2z - k). \quad (107)$$

All nonzero integer samples vanish. If $A_n = |\Phi(n + 1/2)|$, the existing half-integer sign law $\operatorname{sgn} \Phi(n + 1/2) = \epsilon_n$ therefore yields

$$\Phi(z) = \operatorname{sinc}_\pi(2z) + \sum_{n=0}^{\infty} \epsilon_n A_n [\operatorname{sinc}_\pi(2z - (2n + 1)) + \operatorname{sinc}_\pi(2z + (2n + 1))]. \quad (108)$$

This formula is included as a derived synthesis rather than a primary novelty claim, because the latest repository report already develops a broad cardinal-reproduction ladder.

Repo-relative new result

The contour formulas recover the full digit sum and both Thue–Morse normalizations from the zero divisor of Φ . The monodromy formula recovers each finite word from analytic continuation of R_N , while the Abel formula replaces discontinuous Rademacher signs by finite products of harmonic functions.

8 Compact representation atlas

The following table collects the principal forms used in the report. “Baseline” means already present in the audited repository; “new” means derived here and not located in the repository; “synthesis” means a new combination of known ingredients whose components are individually present.

Table 1: Representation atlas.

Object/type	Representation	Status and use
X , random series	$X = \sum_{k \geq 1} 2^{-k} U_k$, $U_k \sim \text{Unif}[-1, 1]$	Baseline; probability model
X , fixed point	$X \stackrel{d}{=} (X + U)/2$	Baseline; source of recurrences
ρ , convolution	$\rho = *_{k \geq 1} (2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]})$	Baseline; compact support and smoothness
ρ_N , spline	$\rho_N = 2^{N(N-1)/2} (N-1)!^{-1} \sum_{j < 2^N} \epsilon_j (x + a_{N,j})_+^{N-1}$	Baseline; finite exact form
ρ , refinement	$\rho(x) = \int_{-1}^1 \rho(2x - u) du$	Baseline; scaling
ρ , derivative	$\rho'(x) = 2[\rho(2x + 1) - \rho(2x - 1)]$	Baseline; differential-difference law
Φ , sinc product	$\prod_{j \geq 0} \text{sinc}_\pi(\xi/2^j)$	Baseline; Fourier image
Φ , zero product	$\prod_{n \geq 1} (1 - \xi^2/n^2)^{1+\nu_2(n)}$	Baseline; dyadic zero divisor
Φ , Bessel series	$J_0(2\pi\xi) + 2 \sum_{m \geq 1} (-1)^m c_{2m} J_{2m}(2\pi\xi)$	New; series–product bridge
$\Phi^{(r)}$	Termwise derivatives of the Bessel series	New; zero-multiplicity sum rules
M , hyperbolic product	$\prod_{k \geq 1} \sinh(t/2^k)/(t/2^k)$	Baseline; moment generating function
M , canonical product	$\prod_{n \geq 1} (1 + t^2/(4\pi^2 n^2))^{1+\nu_2(n)}$	Baseline rearrangement
M , Bessel series	$I_0(t) + 2 \sum_{m \geq 1} c_{2m} I_{2m}(t)$	New; modified-Bessel bridge
μ_n	Triangular rational recurrence (19)	Baseline; exact moments
μ_n	Complete Bell polynomial in Bernoulli cumulants	Baseline; Bell–Bernoulli form
c_{2m}	Finite transform of μ_{2k} , (32)	New; exact probability-Chebyshev coordinates
$G(w)$	$\int (1 - wx)(1 - 2wx + w^2)^{-1} \rho(x) dx$	New; moment generating function in degree
$C(w)$	$\int_{\mathbb{T}} (\zeta + w)(\zeta - w)^{-1} d\nu$	New; Carathéodory transform
$C(w)$, series	$1 + 2 \sum_{m \geq 1} c_{2m} w^{2m}$	New; positive-real disk series
$R(z)$, moments	$\sum_{n \geq 0} \mu_n z^{-n-1}$	Baseline; expansion at infinity
$R(z)$, Laplace	$\int_0^\infty e^{-zt} M(t) dt$	Baseline; integral transform
$R(z)$, Joukowski	$[1 + 2 \sum c_{2m} w(z)^{2m}] / \sqrt{z^2 - 1}$	New; conformal series
$R(z)$, Bessel-Laplace	$\int_0^\infty e^{-zt} [I_0(t) + 2 \sum c_{2m} I_{2m}(t)] dt$	New synthesis

Object/type	Representation	Status and use
$R(z)$, Jacobi CF	$1/(z - \beta_1/(z - \beta_2/(z - \cdots)))$	Baseline; real-line continued fraction
$R(z)$, Schur/CMV	$C(w(z))/\sqrt{z^2 - 1}$, with C a CMV resolvent element	New; unitary operator representation
$\rho(x)$, weighted Cheb	$[1 + 2 \sum c_{2m} T_{2m}(x)]/[\pi \sqrt{1 - x^2}]$	New; boundary Fourier series
$F(u)$, phase series	$1 - \theta/\pi - \pi^{-1} \sum c_{2m} \sin(2m\theta)/m$	New; $\theta = \arccos(2u - 1)$
$F'(u)$	$\Re C(e^{i\theta})/[\pi \sqrt{u(1-u)}]$	New; positive boundary quotient
$I(v)$, phase	$I = (1 + \cos \Theta)/2$, with Θ solving (57)	New; inverse trigonometric representation
$I'(v)$	$\pi \sqrt{I(1-I)}/\Re C(e^{i\Theta})$	New; inverse derivative
F , Fourier	$[e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}]/(2\pi i \xi)$	Baseline
F , Bessel Fourier	Insert the J_{2m} -series for $\Phi(\xi/2)$	New synthesis
I , pushforward	$\int_0^1 e^{-2\pi i \xi I(v)} dv = e^{-\pi i \xi} \Phi(\xi/2)$	Baseline
I , Bessel pushforward	Insert the J_{2m} -series in the preceding formula	New synthesis
OPUC data	Schur recursion with rational $\alpha_{2m} = 0$	New; exact unit-circle coefficients
Jacobi–OPUC	$\beta_{n+1} = \frac{1}{4}(1 - \alpha_{2n-1})(1 + \alpha_{2n+1})$	New application of Geronimus
Toeplitz	$D_N = \prod_{j=0}^{N-2} (1 - \alpha_j^2)^{N-1-j}$	New determinant representation
Szegő product	$\prod (1 - \alpha_n^2) = \exp[(2\pi)^{-1} \int \log W]$	New entropy/product invariant
$s_2(n)$, spectral	$2n - (2\pi i)^{-1} \oint \Phi'/\Phi$	New; full digit sum from zero divisor
ϵ_n , contour	$\exp[\frac{1}{2} \oint \Phi'/\Phi]$	New; signed Thue–Morse
τ_n , contour	$\sin^2[(4i)^{-1} \oint \Phi'/\Phi]$	New; unsigned Thue–Morse
Finite TM word	Normalized $(N-1)$ -st derivative of logarithmic monodromy	New; analytic-continuation encoding
ϵ_n , Abel	Boundary limit of a finite product of H_r -factors	New; smooth harmonic regularization
Φ , cardinal TM	Half-integer Shannon series with coefficients $\epsilon_n A_n$	Synthesis; overlaps repository cardinal program

9 Exact computations and numerical experiments

The file `numerical_experiments.py` is deliberately self-contained. It uses `fractions.Fraction` for exact moments, Chebyshev moments, Jacobi coefficients, and Verblunsky coefficients; `mpmath` for 70-digit analytic checks; and `matplotlib` for the figures. The code follows two independent coefficient routes:

1. ordinary moments $\rightarrow$ exact monic Gram–Schmidt $\rightarrow \beta_n \rightarrow \alpha_n$ through Geronimus;

2. ordinary moments $\rightarrow c_n \rightarrow C(w) \rightarrow \alpha_n$ through a formal exact Schur algorithm.

The two routes agree exactly through α_{15} in the automated assertion. The script also verifies $N_\Phi(n) = 2n - s_2(n)$ for $0 \leq n \leq 256$, the monodromy normalization for $1 \leq N \leq 9$, and all positivity/unit-disk conditions for the computed coefficients.

Table 2: First exact probability–Chebyshev moments.

m	$2m$	c_{2m}
0	0	1
1	2	$\frac{-7}{9}$
2	4	$\frac{227}{675}$
3	6	$\frac{-11237}{297675}$
4	8	$\frac{-8241761}{227721375}$
5	10	$\frac{171586753}{8135075025}$
6	12	$\frac{-472381023339557}{45472425490366875}$

Table 3: First exact nonzero Verblunsky coefficients.

n	α_n	decimal
1	$\frac{-7}{9}$	-0.777777777778
3	$\frac{-17}{25}$	-0.68
5	$\frac{-1849}{3087}$	-0.598963394882
7	$\frac{-3569379}{6492691}$	-0.549753407331
9	$\frac{-445076883510763}{886612917691413}$	-0.501996840594

9.1 Bessel convergence

At 40 even terms, the pointwise errors against a 120-factor product were approximately

ξ	absolute error
0.125	below the 70-digit working threshold
0.7	9.1×10^{-72}
1.75	5.5×10^{-69}
3.2	7.4×10^{-48}

The degradation at $\xi = 3.2$ is expected: a fixed order truncation must resolve more oscillatory Bessel content as $|\xi|$ grows.

Table 4: First exact monic Jacobi recurrence coefficients.

n	β_n	decimal
1	$\frac{1}{9}$	0.111111111111
2	$\frac{32}{225}$	0.142222222222
3	$\frac{619}{3675}$	0.16843537415
4	$\frac{835232}{4640643}$	0.179981955087
5	$\frac{1789043898625}{9272274723657}$	0.192945523288

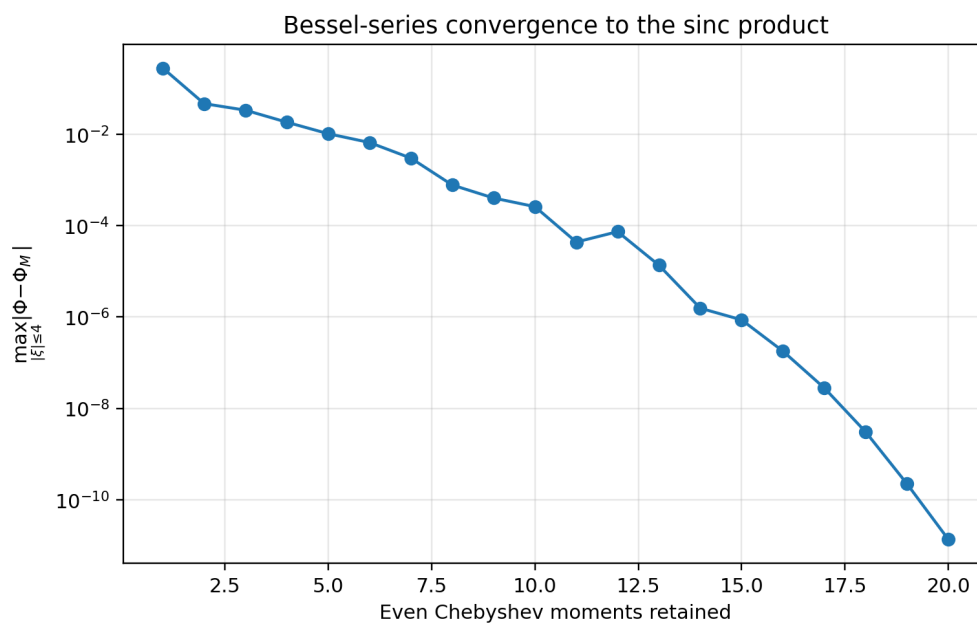


Figure 1: Maximum error on $|\xi| \leq 4$ of the first M even-order terms in (65), measured against a 90-factor sinc product. At $M = 20$, the recorded maximum is approximately 1.35×10^{-11} .

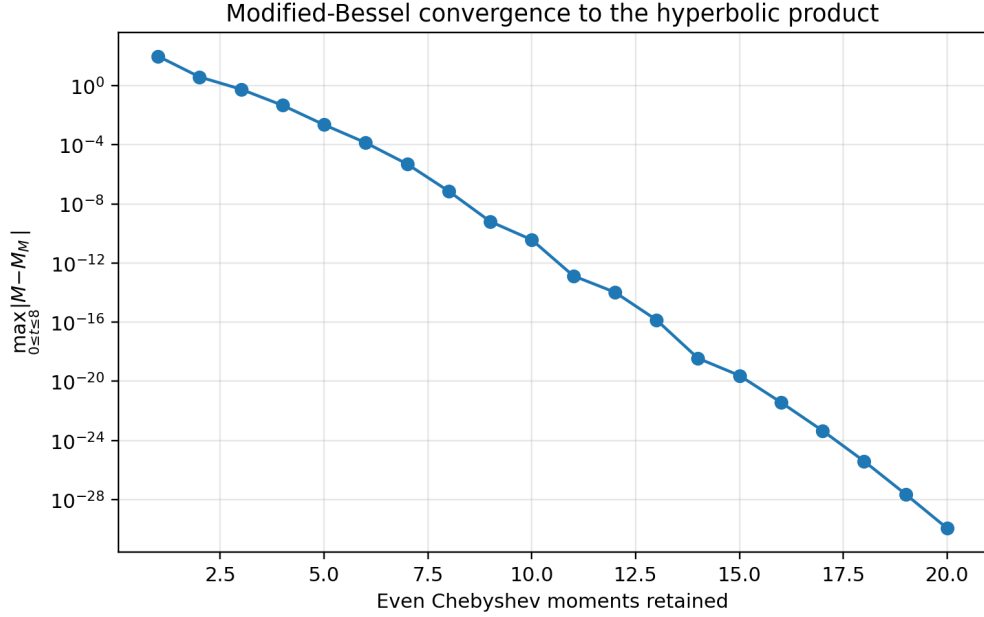


Figure 2: Maximum error on $0 \leq t \leq 8$ of the first M even-order terms in (70), measured against a 90-factor hyperbolic product. At $M = 20$, the recorded maximum is approximately 1.14×10^{-30} .

9.2 Three independent Cauchy-transform evaluations

The Joukowski series, ordinary moment series, and direct Laplace-product integral agree as follows:

z	Joukowski/Chebyshev	moment-series error	Laplace-product error
1.5	0.7040034721763372090	1.71×10^{-22}	5.82×10^{-17}
2.0	0.5148540730138274802	5.75×10^{-33}	1.22×10^{-27}
3.0	0.3375690921956309530	1.21×10^{-47}	3.19×10^{-45}

At $z = 1.5$, the moment series is closer to its unit-circle boundary of convergence and the quadrature tail is harder; the Joukowski coordinate remains well adapted to the slit plane.

9.3 Verblunsky data

The figure is plotted from exact fractions, not from a fitted or floating-point recurrence. The largest fractions contain hundreds of digits; the CSV files preserve their exact numerators and denominators.

9.4 Reproducibility boundaries

The experiments verify identities already proved symbolically; they do not establish the conjectures in Section 10. Product truncation and quadrature errors are not given as formal interval bounds, so their decimal residuals should be read as consistency checks. A natural next step is to port the same recurrences to Arb or MPFI interval arithmetic.

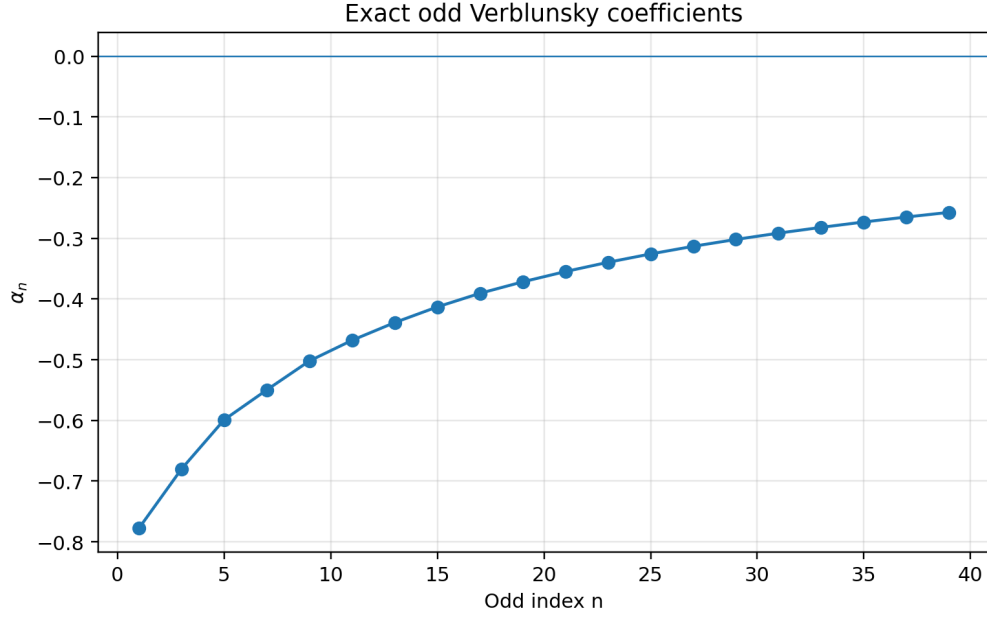


Figure 3: The first twenty exact odd Verblunsky coefficients. All even coefficients vanish. The displayed odd values are negative and increase toward zero, motivating Theorem 10.1; Rakhmanov’s theorem proves convergence to zero but not the observed sign and monotonicity.

10 Conjectures and frontier research program

Conjectural frontier

The statements in this section are unproved unless explicitly described as conditional. They are motivated by exact data, standard OPUC theory, and the known endpoint asymptotics, but they should not be cited as theorems.

Conjecture 10.1 (Sign and monotonicity of odd Verblunsky coefficients). *For every $n \geq 0$,*

$$-1 < \alpha_{2n+1} < \alpha_{2n+3} < 0. \quad (109)$$

Thus the odd subsequence approaches zero strictly from below.

The first twenty odd coefficients support the conjecture. A proof may be accessible through total positivity of Toeplitz minors, a comparison principle for the Schur algorithm, or a special sign-regularity property of the circle weight W . Ordinary positivity of W is not enough; a genuinely stronger structural argument is needed.

Conjecture 10.2 (Logarithmic edge law). *There is a constant $A > 0$ such that*

$$\alpha_{2n-1} \sim -A \frac{\log n}{n}. \quad (110)$$

If the Jacobi edge law has the form

$$\frac{1}{4} - \beta_{n+1} \sim C \frac{(\log n)^2}{n^2}, \quad (111)$$

then the Geronimus relation predicts $A = 2\sqrt{C}$.

Indeed, writing $\gamma_n = -\alpha_{2n-1}$,

$$\frac{1}{4} - \beta_{n+1} = \frac{1}{4}(\gamma_{n+1} - \gamma_n + \gamma_n \gamma_{n+1}).$$

For $\gamma_n \asymp (\log n)/n$, the product term dominates the discrete derivative. This turns the real-line Jacobi edge problem into a unit-circle Schur asymptotic problem.

Conjecture 10.3 (Refined Szegő–Toeplitz asymptotics). *The Toeplitz determinants D_N admit a strong asymptotic expansion whose first correction to the Szegő linear term is governed by the logarithmic endpoint-flatness law of ρ , and hence ultimately by the Lambert– W expansion appearing in the endpoint analysis.*

The weight is positive and smooth but has infinite-order zeros. It lies outside the most naive strictly-positive smooth-weight regime, making it a promising test case for extending Fisher–Hartwig and strong Szegő methods to logarithmically superflat weights.

Conjecture 10.4 (Arithmetic structure of the Szegő constant). *The entropy integral*

$$\mathcal{S}_\rho = \frac{1}{2\pi} \int_0^{2\pi} \log(\pi \rho(\cos \theta) |\sin \theta|) d\theta \quad (112)$$

has a representation in terms of the dyadic cumulant data, perhaps as a rapidly convergent series involving Bernoulli numbers, dyadic zeta factors, or a binary Barnes-type constant.

Conjecture 10.5 (Bessel zero hierarchy has a finite recursive basis). *The derivative cancellation identities obtained from the zero of multiplicity $1 + \nu_2(N)$ at N can be reduced, using Bessel recurrences, to a triangular finite basis whose coefficient matrix has an explicit Thue–Morse or q -binomial factorization.*

Conjecture 10.6 (Quantitative Abel convergence). *For $n \geq 1$ and r sufficiently close to one,*

$$\left| \epsilon_n - \prod_{j=0}^{J_n} H_r \left(\frac{\pi(n + 1/2)}{2^j} \right) \right| \leq C(1 - r) \sum_{j=0}^{J_n} \frac{1}{|\sin(\pi(n + 1/2)/2^j)|} \quad (113)$$

with an absolute constant C , and the reciprocal-sine sum admits a sharp bound in terms of the binary carry structure of n .

10.1 Promising research directions

1. **Rigorous OPUC asymptotics.** Derive the odd Verblunsky asymptotics directly from the superflat weight using nonlinear steepest descent for the OPUC Riemann–Hilbert problem, or via fine spectral theory of the CMV matrix.
2. **Certified coefficient computation.** Combine exact rational recurrences with modular arithmetic and rational reconstruction. Hundreds of α_n and β_n should be feasible without storing enormous intermediate fractions.
3. **Fast inverse Fabius solvers.** Use the phase equation (57) with FFT-evaluated trigonometric sums and safeguarded Newton steps. The positivity of $\Re C$ gives a natural monotonicity certificate.

4. **Entropy and determinant constants.** Numerically evaluate $\mathcal{S}_\rho$, compare it with products of exact $1 - \alpha_n^2$, and use integer-relation searches only after rigorous tail bounds are available.
5. **CMV/Jacobi scattering.** Treat the Rvachev Jacobi and CMV matrices as compact or slowly varying perturbations of the free models. The endpoint-flatness law should appear in reflection coefficients and spectral phase shifts.
6. **Base- b and q -Rvachev lifts.** For generalized products with scales b^{-k} or exponent sequences, construct the analogous circle lift, identify the Carathéodory function, and compare rationality/algebraicity of its Schur parameters.
7. **Monodromy categories.** Replace the binary finite spline by finite q -binomial or digit-weighted splines. Their logarithmic resolvents should carry a monodromy representation of the corresponding automatic word.
8. **De Branges and Hermite interpolation.** The integer zero divisor has unbounded multiplicity. Build a complete interpolation basis adapted to the values of $\Phi^{(r)}(n)$ for $0 \leq r \leq \nu_2(n)$, rather than ordinary simple-node Shannon sampling.
9. **Bessel sum-rule algebra.** Organize (69) and its higher derivatives as identities in the Bessel differential algebra; search for finite closed combinations at powers of two.
10. **Mellin matching.** Match the unit-circle phase representation to the repository’s Mellin/Newton and Lambert- W endpoint expansions, obtaining uniform approximations from the center to the boundary.
11. **Formal verification.** The moment recurrence, exact Chebyshev transform, Carathéodory series, and contour zero count are suitable for Lean formalization. The analytic boundary and Szegő portions would require more functional-analysis infrastructure.
12. **Interval numerics.** Recompute the figures with ball arithmetic, certify the Schur parameters and Bessel identities, and establish explicit truncation bounds for the phase series and Abel product.

11 Conclusion

The audited repository already contains a remarkably broad representation theory: random series, splines, Thue–Morse coefficients, sinc and canonical products, exact moments, Bernoulli/Bell structure, resolvents, logarithmic finite transforms, determinant and continued-fraction forms, multiresolution expansions, integral transforms, inverse-function identities, and endpoint asymptotics. The main advance here is therefore not another isolated formula, but a new organizing bridge.

The probability–Chebyshev moments c_n lift the Rvachev law to the circle. From that lift, one analytic function C simultaneously yields the density, the Fabius phase series, the inverse phase equation, the Joukowski Cauchy transform, Schur and CMV data, and Toeplitz products. The same coefficients turn the infinite sinc and hyperbolic products into ordinary and modified Bessel series. On the arithmetic side, the integer zero divisor of Φ recovers the complete binary digit sum by the argument principle; finite logarithmic resolvents recover Thue–Morse words by monodromy; and Abel regularization gives a smooth finite-product representation.

These bridges place the Rvachev–Fabius system at an intersection of probability, Fourier analysis, special functions, OPUC, operator theory, complex monodromy, automatic sequences, and dyadic asymptotics. The conjectures above are concrete enough for exact computation and theoretical attack, yet broad enough to connect several currently separate campaigns in the repository.

A Proof details and convergence notes

A.1 Smoothness of the circle weight

Near $\theta = 0$, $1 - \cos \theta \asymp \theta^2$. Since ρ and all its derivatives vanish at $x = 1$, the composite $\rho(\cos \theta)|\sin \theta|$ and all its derivatives tend to zero at the origin. The same holds at π and 2π . This justifies the C^∞ periodic extension used in Theorem 4.5, hence the superalgebraic decay of c_n .

A.2 Local uniform convergence of Bessel series

For $|z| \leq R$, the standard bound $|J_n(z)| \leq (R/2)^n/n! e^{R^2/4}$ is sufficient. Since $|c_n| \leq 1$, the series in (65) converges uniformly on compact sets. An analogous factorial-in-order bound for I_n proves local uniform convergence of (70). Termwise differentiation follows from Weierstrass convergence.

A.3 Log-integrability of the circle weight

The established endpoint analysis of $\rho(1 - h)$ has logarithm growing on the scale of a quadratic or mildly corrected quadratic polynomial in $\log(1/h)$. With $h \asymp \theta^2$, any finite power of $\log(1/\theta)$ is integrable near zero:

$$\int_0^\varepsilon |\log^k \theta| d\theta < \infty.$$

The extra factor $|\sin \theta|$ contributes only $\log |\theta|$. Therefore $\log W \in L^1$, as required by Szegő's theorem.

A.4 A direct expression for the circle measure

For any continuous h on $\mathbb{T}$,

$$\int_{\mathbb{T}} h(\zeta) d\nu(\zeta) = \int_{-1}^1 \frac{h(e^{i \arccos x}) + h(e^{-i \arccos x})}{2} \rho(x) dx. \quad (114)$$

This formula is often the fastest route for transporting real-line identities to the circle.

B Reproducibility map

File	Purpose
<code>numerical_experiments.py</code>	Exact moments, c_n , Jacobi and Verblunsky coefficients; Bessel/product, Cauchy, spectral, Abel, and monodromy checks; figure and table generation.
<code>numerical_results.txt</code>	Human-readable verification transcript with decimal residuals.
<code>chebyshev_moments.csv</code>	Exact numerators, denominators, and decimal values of c_n .

File	Purpose
<code>jacobi_coefficients.csv</code>	Exact β_n .
<code>schur_parameters.csv</code>	Exact α_n , including structural zeros.
<code>generated_tables.tex</code>	Tables inserted into this report.
<code>corpus_audit.txt</code>	Audit method, overlap exclusions, and novelty language.
<code>README.txt</code>	Reproduction and compilation instructions.

The exact coefficient computation starts from only (19); all other rational quantities are consequences. This makes the computational chain auditable and suitable for independent reimplementations.

References

- [1] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction/docs*, living research repository, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.
- [2] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 5 (1966), 173–174.
- [3] V. L. Rvachev, On a function with compact support that is infinitely differentiable, original work introducing the compactly supported “up” function; bibliographic and translated versions are catalogued in the repository’s historical source files.
- [4] NIST Digital Library of Mathematical Functions, Chapters 10 and 11: Bessel functions and generating/Jacobi–Anger expansions, <https://dlmf.nist.gov/10>.
- [5] B. Simon, *Orthogonal Polynomials on the Unit Circle, Part 1: Classical Theory*, American Mathematical Society Colloquium Publications, vol. 54.1, 2005.
- [6] B. Simon, OPUC on one foot, in *Bull. Amer. Math. Soc.* and related expository notes on Carathéodory functions, Schur parameters, CMV matrices, and Geronimus relations.
- [7] G. Szegő, *Orthogonal Polynomials*, American Mathematical Society Colloquium Publications, fourth edition, 1975.
- [8] B. Simon, *Orthogonal Polynomials on the Real Line*, American Mathematical Society Colloquium Publications, vol. 54.4, 2005.
- [9] E. T. Whittaker, On the functions which are represented by the expansions of the interpolation theory, *Proceedings of the Royal Society of Edinburgh* 35 (1915), 181–194.
- [10] C. E. Shannon, Communication in the presence of noise, *Proceedings of the IRE* 37 (1949), 10–21.
- [11] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [12] N. I. Akhiezer, *The Classical Moment Problem and Some Related Questions in Analysis*, Oliver & Boyd, 1965.